

TheFacebook: Topic 2 Data Structure Project Report

WIA1002 Data Structure Group Assignment

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1. Project Introduction

TheFacebook is a Java-based social networking application developed for Topic 2 of the WIA1002 Data Structure group assignment. The project simulates an early social media platform where users can create accounts, log in, edit their profiles, search for other users, send and respond to friend requests, view mutual friends, receive friend recommendations, display first-, second-, and third-degree friends, create group conversations, chat with friends, and generate a simple analysis report.

The main purpose of the project is to apply data structures in a meaningful software system. A social network is suitable for this topic because user accounts need efficient searching, friendships need graph representation, friend requests need organized processing, group and chat records need dynamic collections, and session history can be represented using a LinkedList.

The application was first built as a console program and later extended with a simple Java Swing graphical user interface. Both interfaces use the same service layer and data files, so the GUI does not replace the original logic. Instead, it provides a more user-friendly way to demonstrate the same functions.

2. Project Objectives

- To design a simple social media system using object-oriented programming.
- To apply suitable data structures such as graph, queue, stack, ArrayList, LinkedList, and custom hash table.
- To store user, friendship, friend request, private chat, and group chat data permanently using CSV files.
- To implement the basic requirements of Topic 2, including registration, login, account editing, friend searching, mutual friends, friend requests, and mock data.
- To include extra features such as GUI, password hashing, data analysis, enhanced networking, friend chat, group chat, browsing history navigation, and degree-based friend display.

3. System Overview

The system is organized into packages so that each part of the program has a clear responsibility. The model package stores data objects, the service package controls the business logic, the datastructures package stores self-built or selected data structures, and the ui package handles user interaction.

Class	Main Responsibility
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Main	Loads data and starts either the GUI or the console interface.
User	Stores user profile information, password hash, hobbies, career history, and role.
Role	Defines USER and ADMIN roles.
FriendRequest	Represents one friend request from one user to another user.
Message	Stores one chat message with sender ID, timestamp, and text content.
FriendConversation	Stores one-to-one chat records between two friends.
GroupConversation	Stores group ID, group name, owner, members, and group messages.
SocialNetwork	Controls core functions such as login, registration, search, friend requests, recommendations, mutual friends, degree friends, chat, groups, reporting, and deletion.
CsvStorage	Handles reading and writing users, friendships, requests, private messages, groups, group messages, and reports.
PasswordUtil	Hashes passwords using SHA-256.
MyHashTable	Custom generic hash table used for fast lookup.
FriendGraph	Stores friendships and performs graph-related operations including BFS degree search.
BrowsingHistory	Original linked-node history used by the console traceback function.
LinkedListBrowsingHistory	GUI browsing history implemented with Java LinkedList, supporting history view, back, forward, and clear.
ConsoleUI	Provides the original menu-based console interface.
TheFacebookGUI	Provides the Swing graphical interface and connects buttons/dialogs to service methods.

4. Basic Features and Implementation Methods

4.1 Data Storage

The program stores data in CSV files inside the data folder. `users.csv` stores account information, `friendships.csv` stores friendship edges, `requests.csv` stores pending friend requests, `friend_messages.csv` stores private chat records, `groups.csv` stores group information, and `group_messages.csv` stores group chat records. `CsvStorage` uses file reading and writing to make sure persistent data remains after the program closes.

4.2 Login, Registration, and Account Setup

The `register` method in `SocialNetwork` checks password confirmation, email or phone format, and duplicate username, email, or phone number. Passwords are hashed before storage. The `login` method searches by email first, then by phone number, and compares the hashed password.

4.3 Edit Account

Users can edit profile fields such as name, birthday, address, gender, relationship status, hobbies, and career history. Hobbies are stored in an `ArrayList` because the list can grow dynamically. Career history is stored in a `Stack` so the newest record can be shown first.

4.4 Add New Friends

A user can send a friend request to another user. Pending requests are stored in a `Queue` because requests can be reviewed in first-in-first-out order. When a request is accepted, `FriendGraph` adds an undirected friendship edge between the two users.

4.5 User Access Control

The `Role` enum separates `USER` and `ADMIN` accounts. Admin users can access the delete user function. This demonstrates basic access control using object-oriented programming.

4.6 Search for Friends

Users can search by name, username, ID, email, or contact number. Search results are placed in an `ArrayList` and sorted by name before being displayed.

4.7 View Account

The `publicProfile` method in `User` formats important profile information, including name, username, gender, age, address, relationship status, hobbies, career history, friend count, and role.

4.8 Mutual Friends

Mutual friends are found by getting the friend lists of two users from the graph and comparing them. If the same user ID appears in both lists, that user is a mutual friend.

4.9 Browsing History

LinkedListBrowsingHistory uses Java's LinkedList to store actions for the current login session. It supports recording visited content, viewing the full history list, moving back to the previous record, moving forward to the next record, and clearing all records when the user logs out.

4.10 Friend Chat

FriendConversation represents one-to-one conversation between two friends. A private message can only be sent if both users already have a friendship edge in FriendGraph. Each message is stored as a Message object with sender ID, sending time, and text content. Private chat records are saved in friend_messages.csv.

4.11 Group Conversation

GroupConversation represents a group chat. It stores a group ID, group name, owner ID, member ID list, and group messages. Member IDs and messages are stored in ArrayLists because they need dynamic growth and simple ordered traversal. Groups are indexed in SocialNetwork by MyHashTable<String, GroupConversation> for fast lookup by group ID.

4.12 Degree Friends

FriendGraph uses breadth-first search to find first-degree, second-degree, and third-degree connections. First-degree users are direct friends, second-degree users are friends of friends, and third-degree users are one more level away. The GUI displays these three groups through the Degree Friends button.

4.13 Mock Data

The data files contain 30 normal users and 2 admin users. Additional CSV files also support friend messages, group records, and group messages. This allows the system to be tested without manually registering many accounts.

5. Data Structures Used

Data Structure	Used In	Reason for Selection
Custom Hash Table	usersById, usersByEmail, usersByPhone, usersByUsername, groupsById	Provides fast lookup and avoids direct use of java.util.HashMap.
Graph	FriendGraph	Friendship connections are naturally represented as users connected by edges. BFS supports recommendations and first-, second-, and third-degree display.
Queue	Friend requests	Friend requests can be handled in first-in-first-out order.

ArrayList	Search results, friends, hobbies, recommendations, group members, private messages, and group messages	Suitable for dynamic lists that are often traversed and displayed in order.
Stack	Career history	The newest career or study record can be pushed and shown first.
LinkedList	LinkedListBrowsingHistory	Browsing history is a sequence that supports moving backward, moving forward, removing old forward history, and clearing on logout.

6. Extra Features

Extra Feature	Implementation Details
Graphical User Interface	TheFacebookGUI.java uses Java Swing components such as JFrame, JPanel, JButton, JTextField, JPasswordField, JOptionPane, JScrollPane, and JTextArea.
Password Hashing	PasswordUtil uses SHA-256 so passwords are not stored in plaintext.
Data Analysis	SocialNetwork can generate analysis_report.txt with total accounts, admins, normal users, friendship count, pending requests, average age, and gender count.
Enhanced Networking	FriendGraph uses BFS to calculate connection degree up to the third degree and support friend recommendations.
Degree Friends	The Degree Friends GUI button displays first-degree, second-degree, and third-degree users separately.
Friend Chat	FriendConversation and Message store one-to-one messages between friends. Messages are saved in friend_messages.csv.
Group Conversation	GroupConversation stores group details, members, and messages. Group data is saved in

	groups.csv and group_messages.csv.
Browsing History Navigation	LinkedListBrowsingHistory stores current-session GUI actions and supports History, Back, Forward, and Logout clear.
Add Friend	Users are allowed to send friend requests to registered users, and accepted requests become graph edges.

7. Member Roles and Contributions

The following table divides the work among five members. The member names can be replaced with the actual names before submission. Each class is assigned to the member who is mainly responsible for writing and explaining it.

Member	Main Role	Classes / Files Completed	Detailed Contribution
Member 1 Name: Dong hanyu (20251960202)	Project leader and core service developer	SocialNetwork.java, ConsoleUI.java, FriendRequest.java, FriendConversation.java	Designed the program flow, connected storage and UI layers, implemented registration, login, update profile, search, friend request handling, mutual friends, recommendations, admin deletion, report generation, mock data loading, and private chat service methods.
Member 2 Name:Chen Linkai (20251960250)	Data model developer	User.java, Role.java, Message.java, GroupConversation.java	Created the user account object, role enum, message object, group conversation object, CSV conversion methods, age calculation, public

			profile display, hobbies ArrayList, and career Stack.
Member 3 Name:Wang Jingyu (20251960115)	Data structure developer	MyHashTable.java, FriendGraph.java, BrowsingHistory.java, LinkedListBrowsingHistory.java	Implemented the custom generic hash table, adjacency-list graph, BFS connection degree, mutual friend logic, recommendation sorting, linked-node traceback history, and LinkedList browsing history.
Member 4 Name:Li Zhongwu (20251960283)	Storage and security developer	CsvStorage.java, PasswordUtil.java, data/*.csv	Implemented CSV reading and writing for users, friendships, requests, private messages, groups, group messages, and reports; handled data folder management, SHA-256 password hashing, and mock data checking.
Member 5 Name:Hu Shinuo (20251960203)	User interface and testing developer	Main.java, TheFacebookGUI.java	Built the menu-based console interface, added the Swing GUI, connected buttons and dialogs to SocialNetwork methods, tested main user flows, and added GUI access to chat, group, history, and degree friend functions.

8. Problems Faced and Solutions

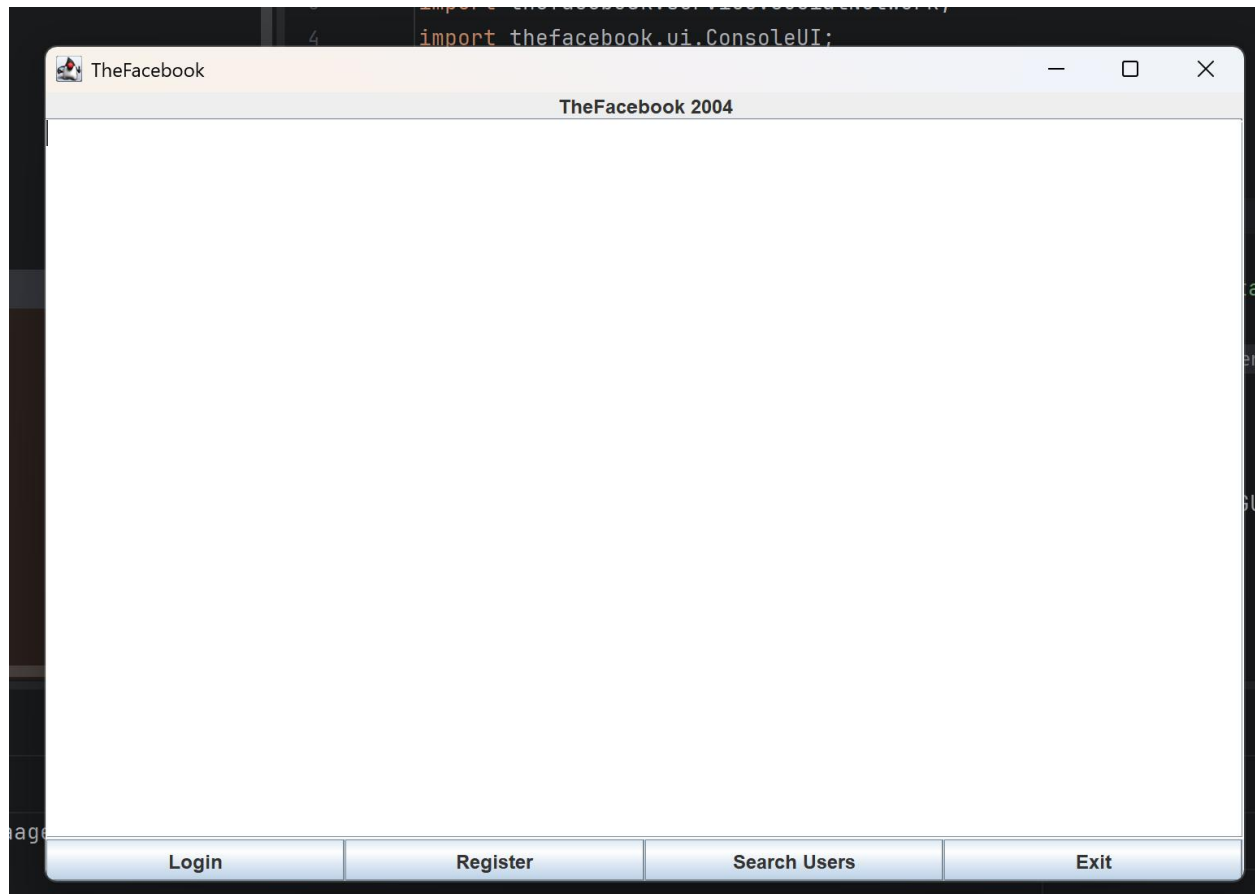
Problem	Solution
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Direct use of HashMap was not allowed by the assignment.	A custom generic MyHashTable class was implemented using buckets and linked entries. It supports put, get, remove, keys, values, clear, and resize.
Friend relationships are difficult to manage using only arrays or lists.	A graph was used. Each user is a node and each friendship is an undirected edge. This makes mutual friends and connection degree easier to implement.
Data should not disappear after the program closes.	CsvStorage was created to save users, friendships, requests, private messages, group records, and group messages into CSV files, then load them again when the program starts.
Passwords should not be stored as plaintext.	PasswordUtil hashes passwords with SHA-256 before saving them to users.csv.
The console interface was not attractive enough as a social media application.	A simple Swing GUI was added. It uses basic components so it is easy to understand and suitable for a first-year project.
The GUI and console could have duplicated the same logic.	Both interfaces call the same SocialNetwork methods where possible, so the business logic remains in one place.
Friend recommendations and degree display need to consider indirect connections.	Breadth-first search is used in FriendGraph to explore up to third-degree connections.
Chat and group data need ordered dynamic storage.	ArrayList is used for group members and messages because records can grow dynamically and are displayed in order.
Browsing history needs back and forward movement.	LinkedListBrowsingHistory uses Java LinkedList with a current index to support visit, show history, back, forward, and clear.
The program needed enough test users.	The seed data and CSV files contain 30 normal users and 2 admins, satisfying the mock data requirement.

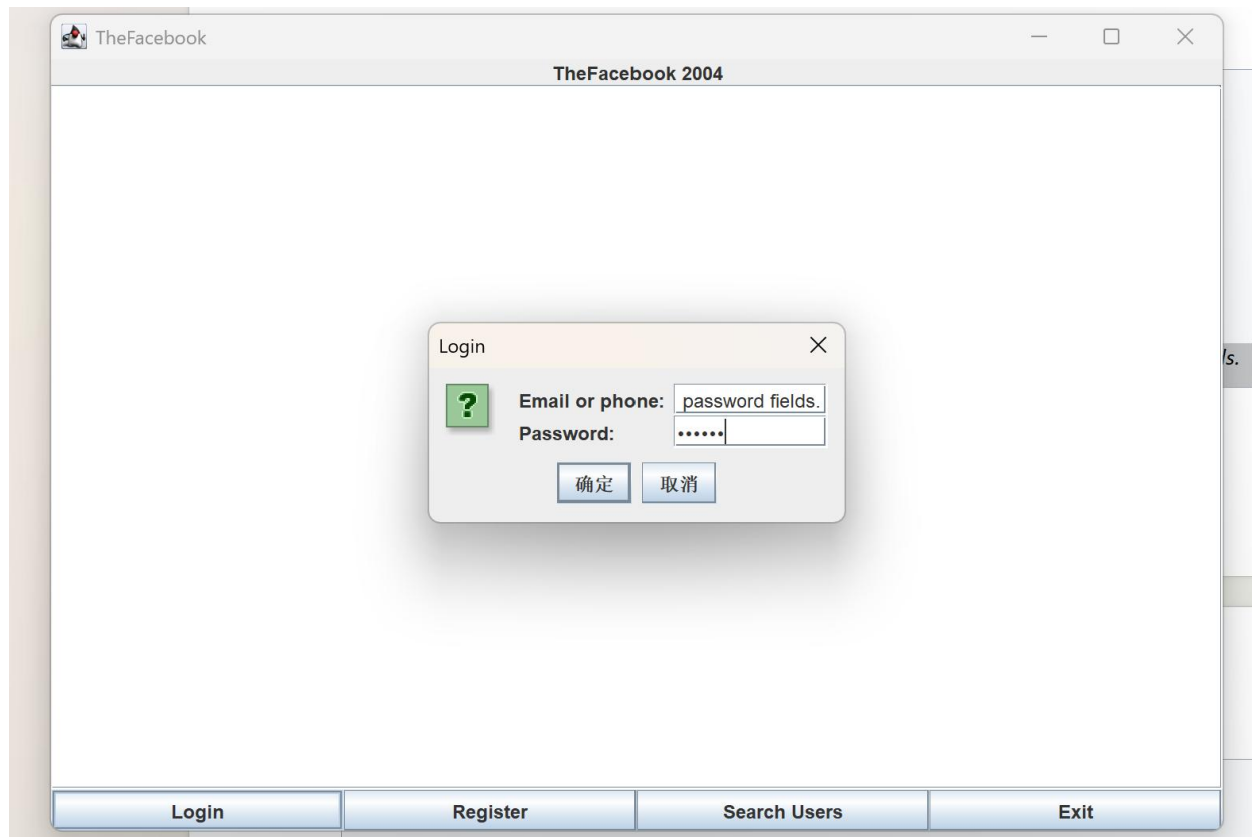
9. Program Screenshots

Please insert the actual screenshots into the blank boxes below before final submission. Each screenshot should show the full application window clearly.

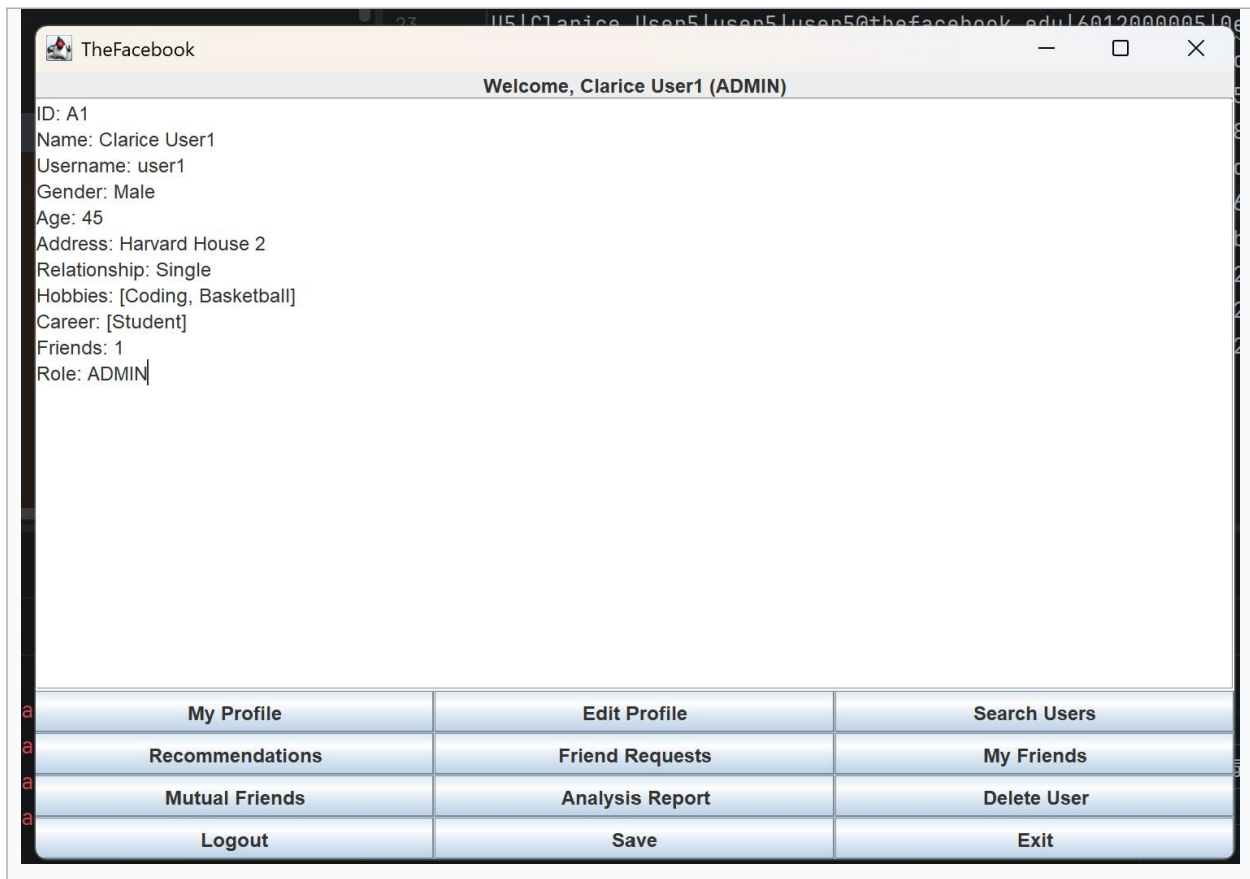
Screenshot 1: GUI Start Page



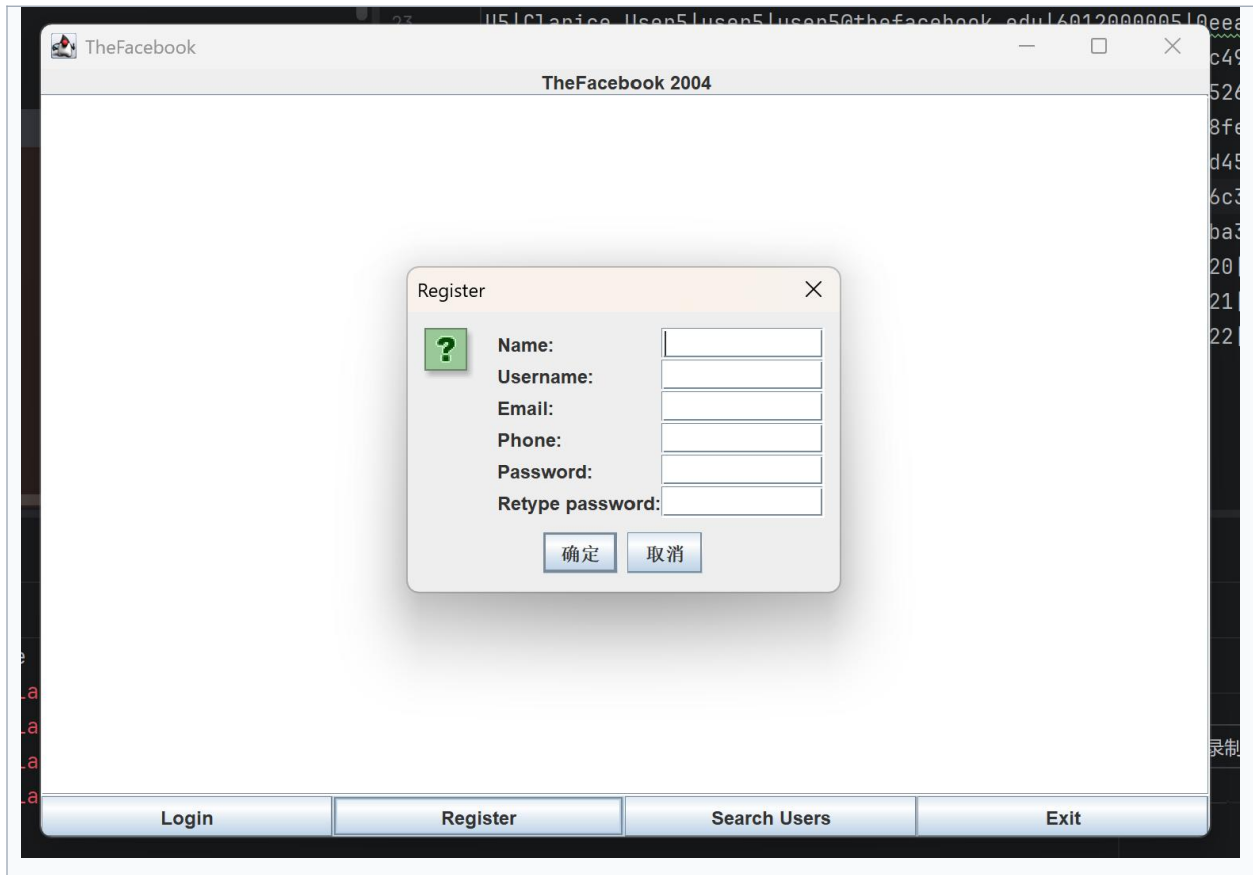
Screenshot 2: Login Dialog



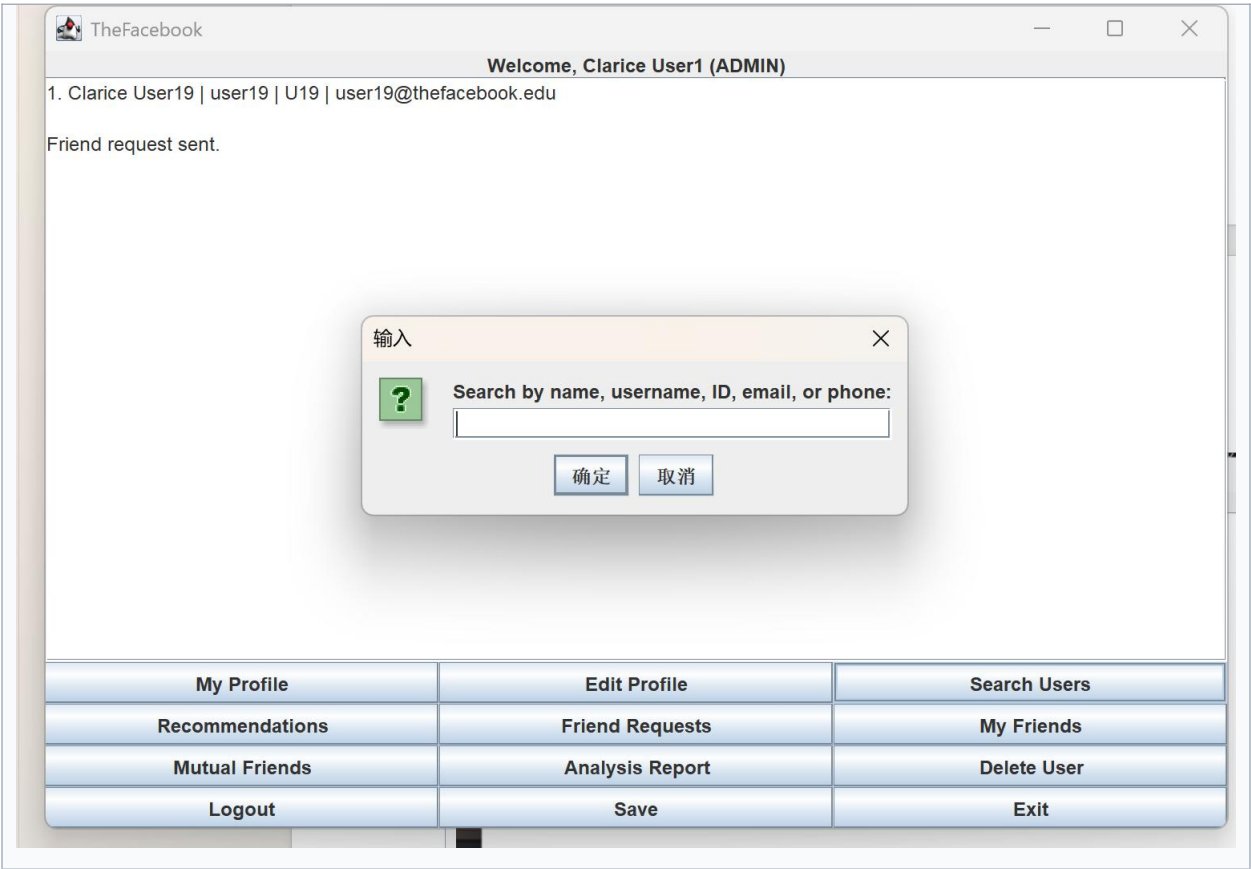
Screenshot 3: User Home Page



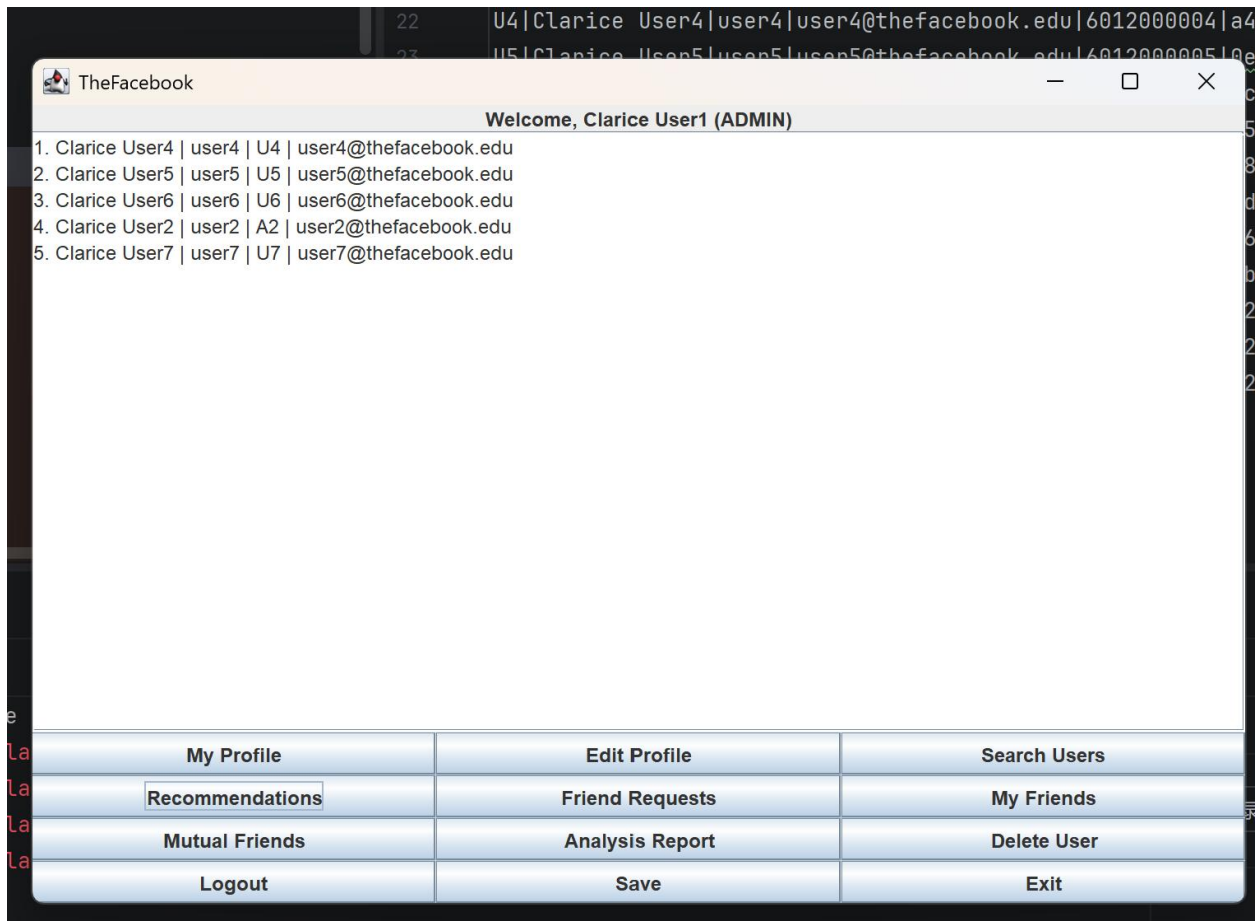
Screenshot 4: Register Function



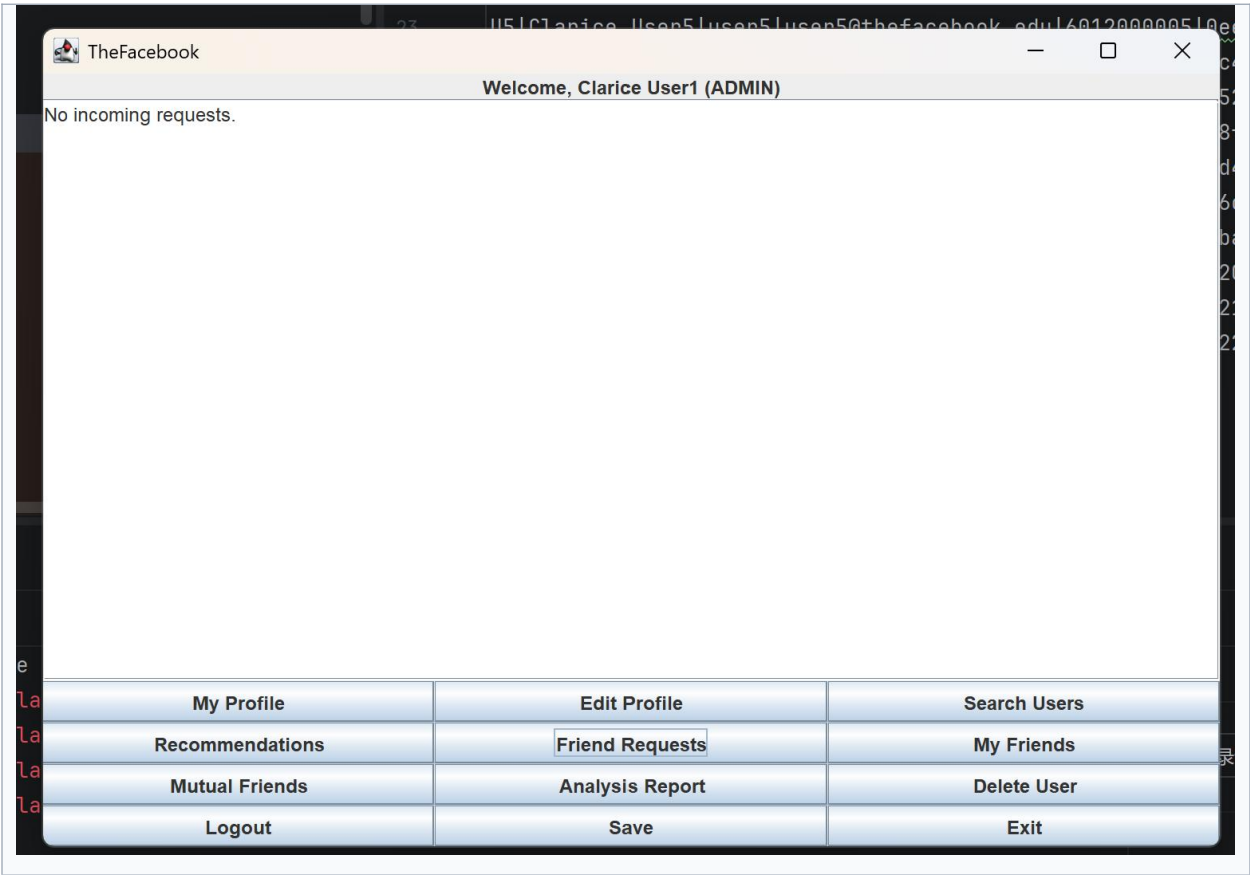
Screenshot 5: Search Users Function



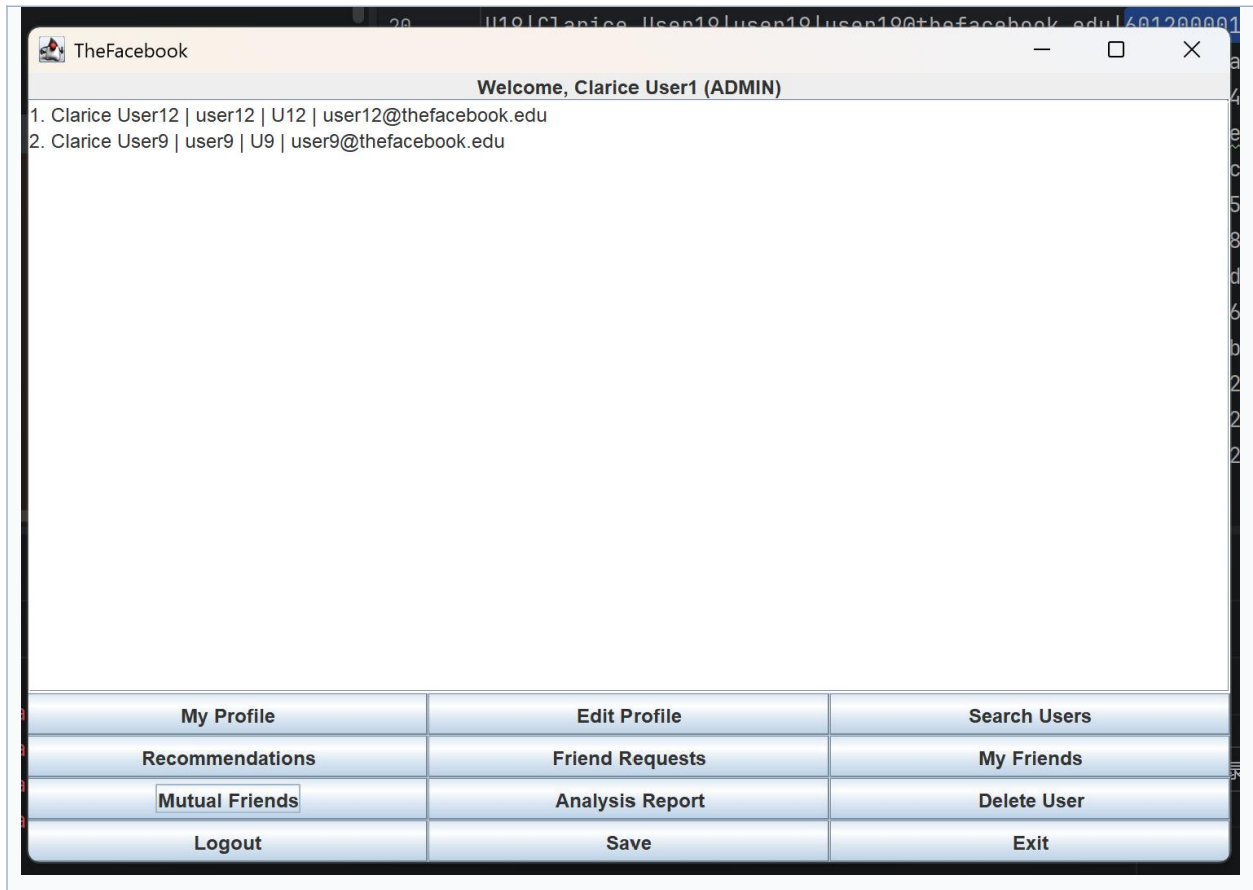
Screenshot 6: Friend Recommendations



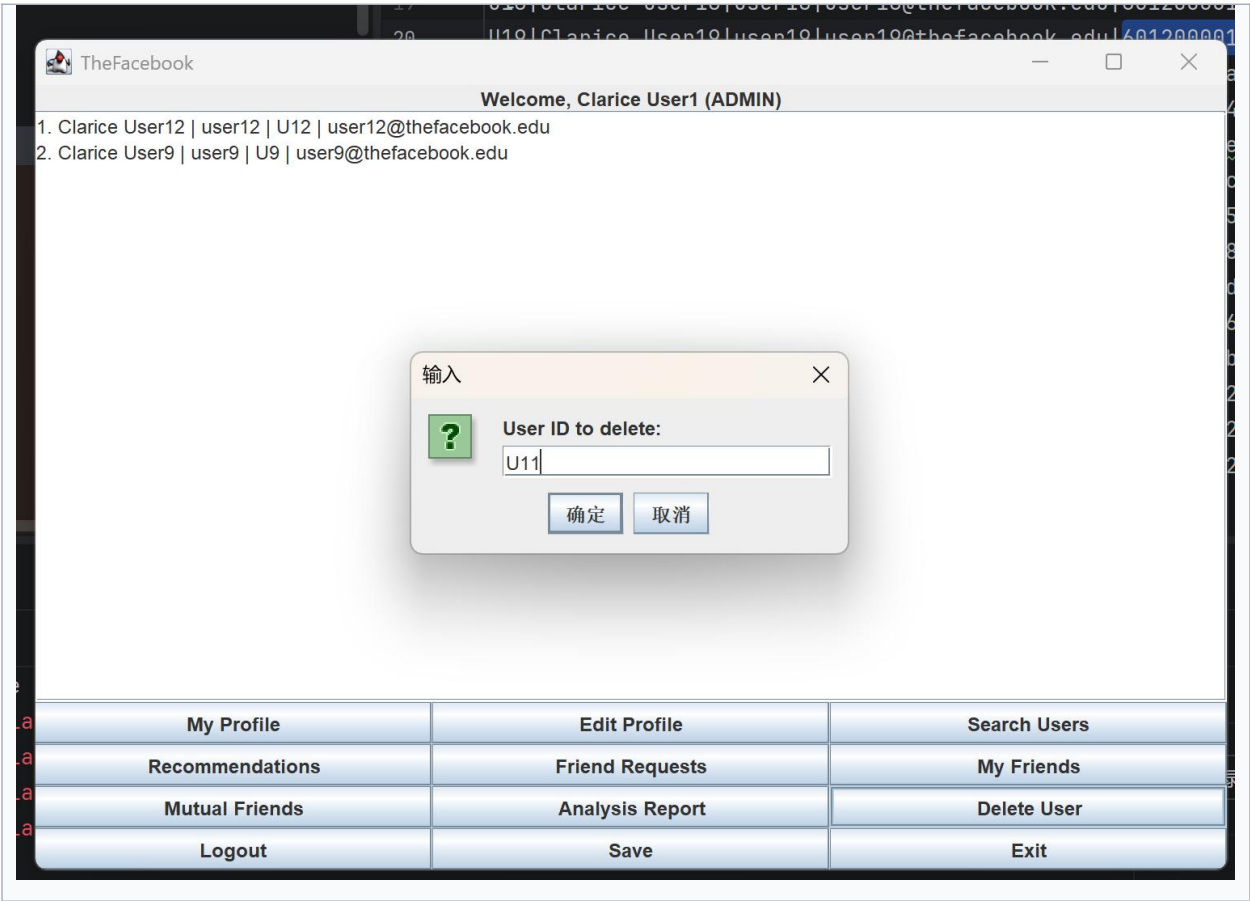
Screenshot 7: Friend Requests



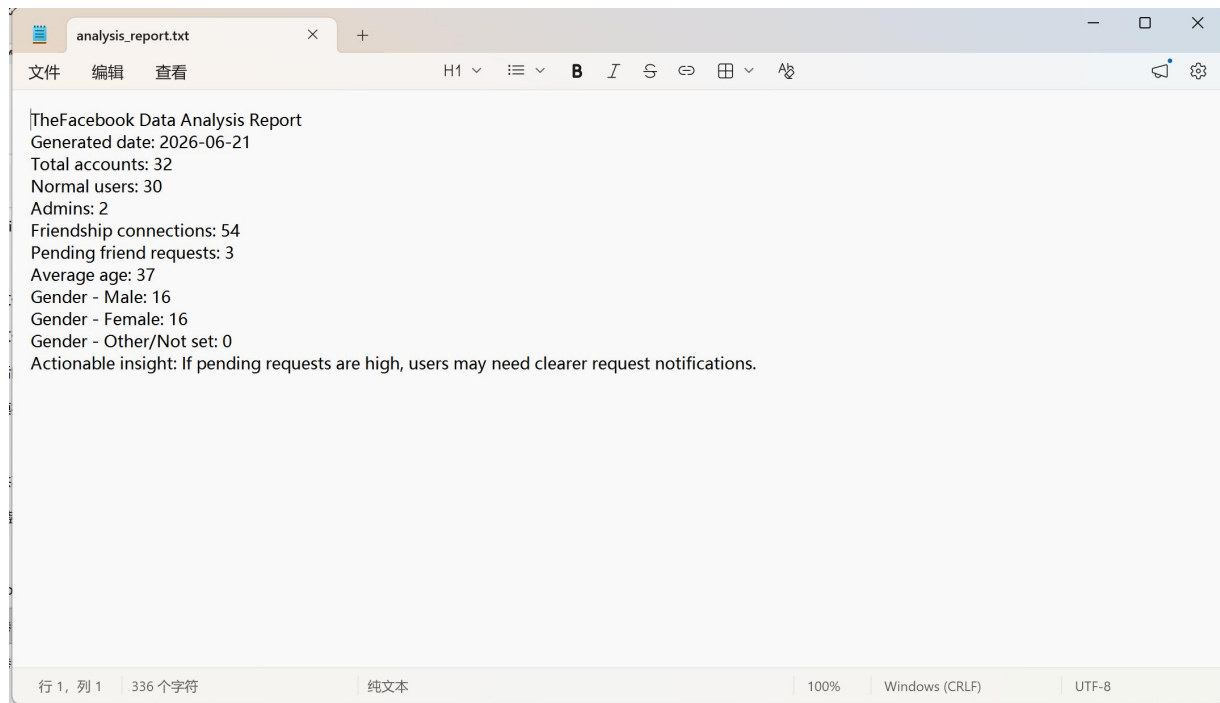
Screenshot 8: Mutual Friends



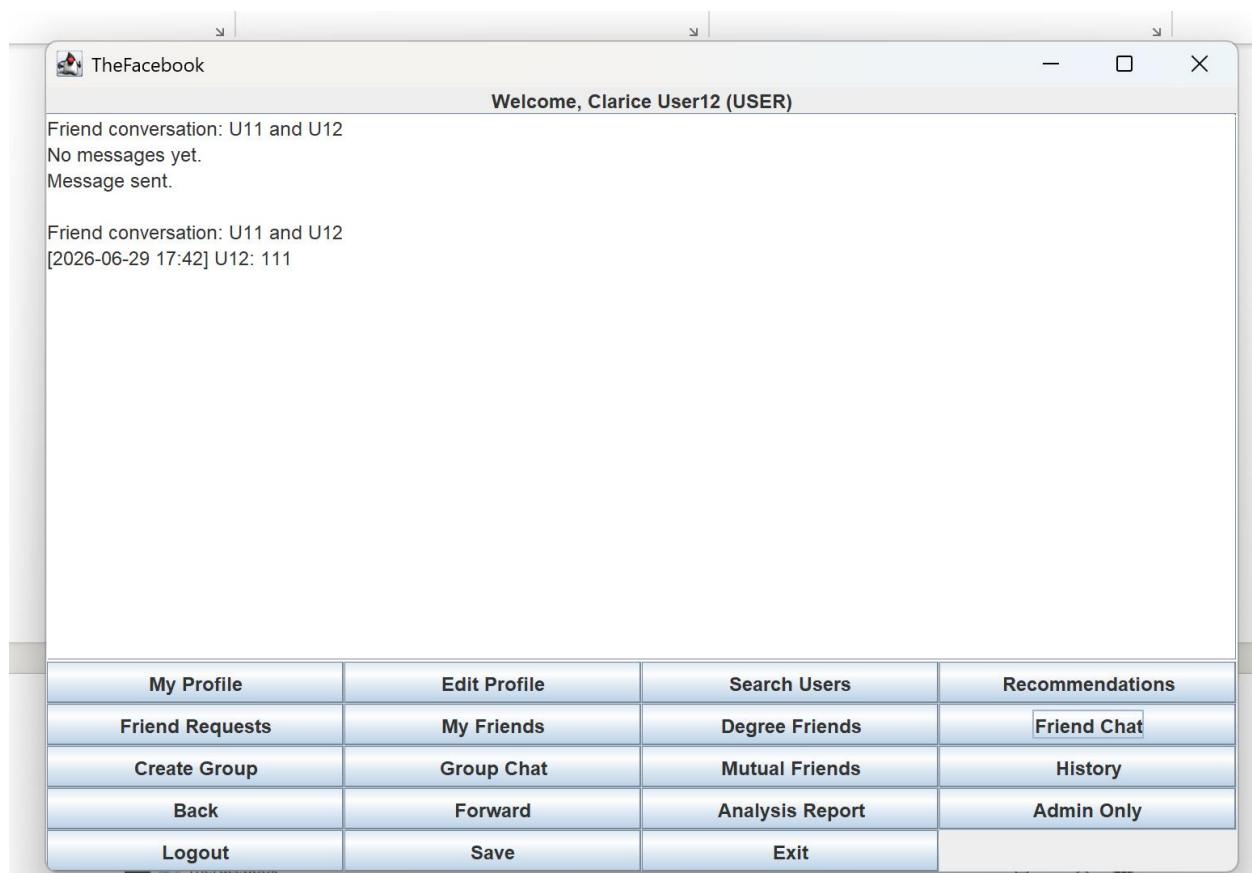
Screenshot 9: Admin Delete User



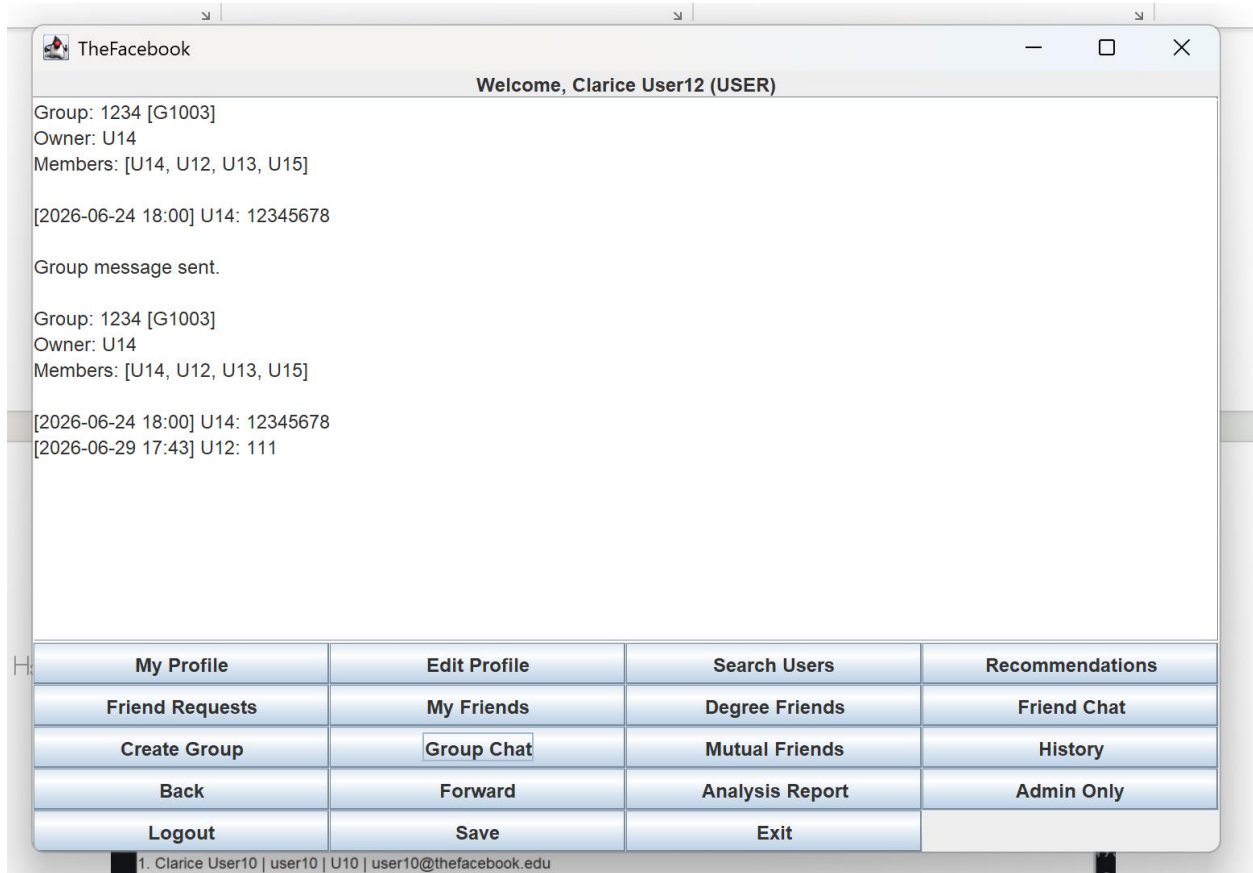
Screenshot 10: Data Analysis Report



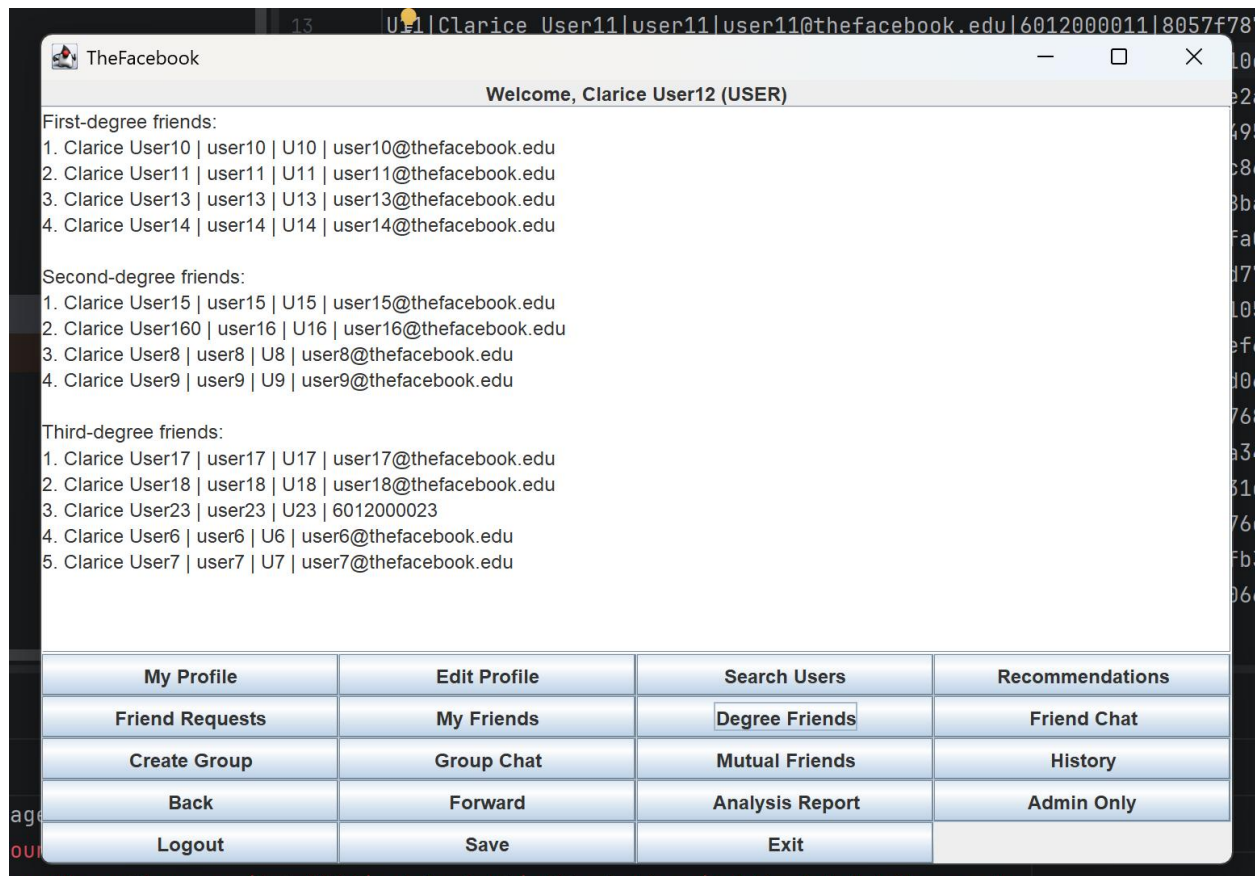
Screenshot 11: Friend Chat



Screenshot 12: Group Chat



Screenshot 13: Degree Friends



10. Testing Summary

- The Java source files were compiled successfully using javac after adding chat, group conversation, LinkedList browsing history, and degree friend display.
- The console version was tested by running the program with the console argument.
- The GUI version was tested by opening the Swing window and checking the main buttons and dialogs, including Friend Chat, Create Group, Group Chat, History, Back, Forward, and Degree Friends.
- The data folder was checked and contains 32 mock users: 30 normal users and 2 admin users.
- The CSV files in the data folder were checked, including users.csv, friendships.csv, requests.csv, friend_messages.csv, groups.csv, and group_messages.csv.

11. Limitations and Future Improvements

- The GUI is intentionally simple. It can be improved with tables, better layout, profile pictures, and a richer chat window.

- The browsing history function stores session actions and supports back/forward movement, but it can be improved to reopen the exact previous profile or content page instead of only displaying the record text.
- Editing username, email, or phone number can be improved with duplicate checking.
- Future features can include fuzzy search, privacy settings, file or image messages, group member management, and a real database such as MySQL.

12. Conclusion

TheFacebook demonstrates how data structures can support a social networking application. The project uses a custom hash table for fast user and group lookup, a graph for friendship relationships and degree-based connections, a queue for friend requests, an ArrayList for dynamic collections and chat messages, a stack for career history, and a LinkedList for browsing history. The Swing GUI improves usability and now supports friend chat, group chat, browsing history navigation, and degree friend display. Overall, the program covers the major basic requirements of Topic 2 and includes meaningful extra features that can be explained clearly during presentation.